EXP:5	EDA – DATA VISUALIZATION

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

data = {
   'Category': ['A', 'B', 'C', 'D'],
   'Values': [3, 7, 5, 4]
}

df = pd.DataFrame(data)

print("Dataset:")
display(df)

x = np.linspace(0, 10, 100)
y = np.sin(x)

plt.plot(x, y, label='Sine Wave')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('Line Chart')
plt.legend()
```

```
plt.show()


plt.bar(df['Category'], df['Values'])
plt.xlabel('Category')
plt.ylabel('Values')
plt.title('Bar Chart')
plt.show()

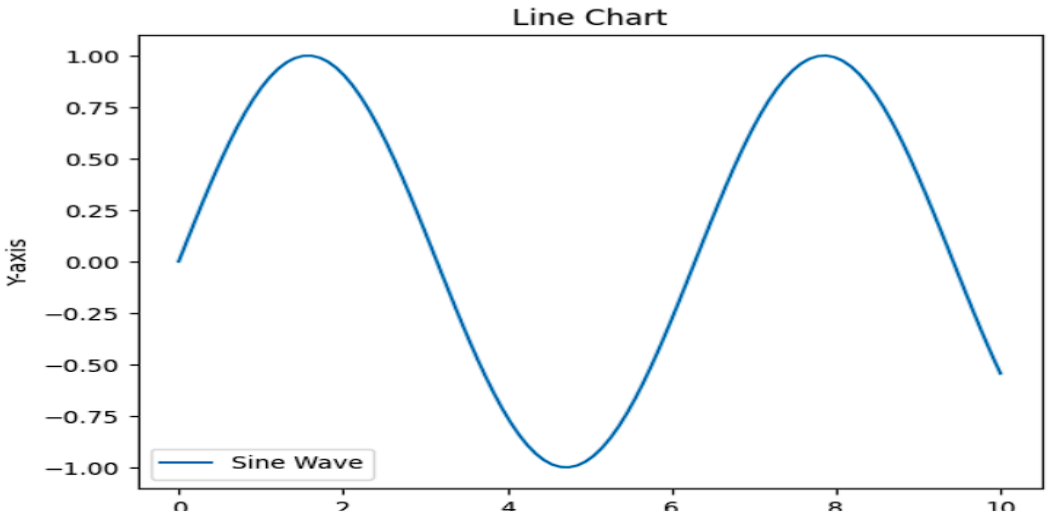

random_data = np.random.randn(1000)

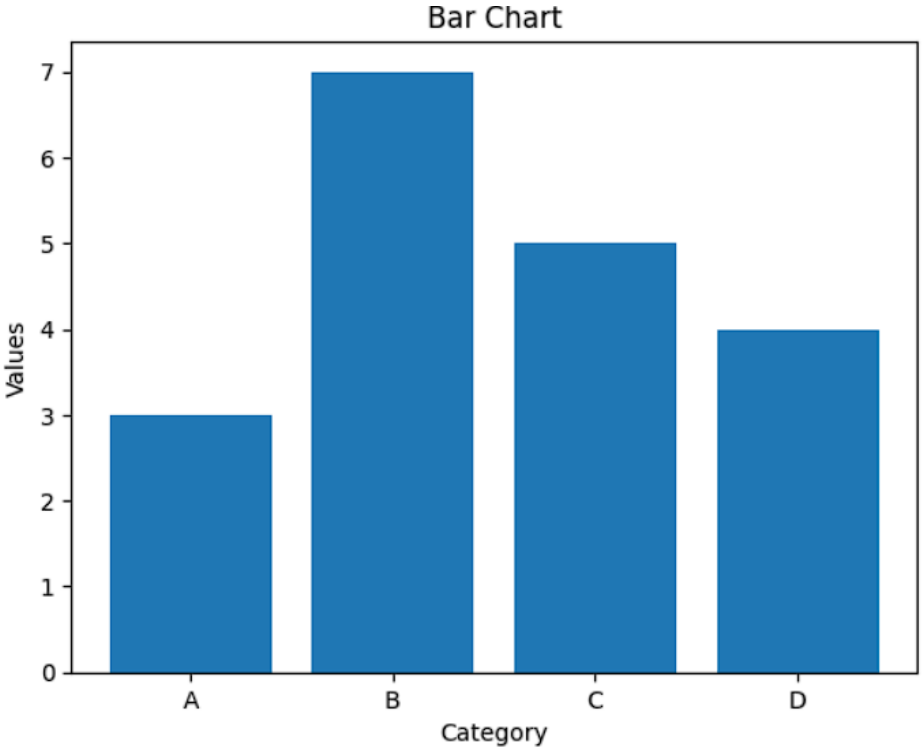

plt.hist(random_data, bins=20, edgecolor='black')
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.title('Histogram')
plt.show()
```

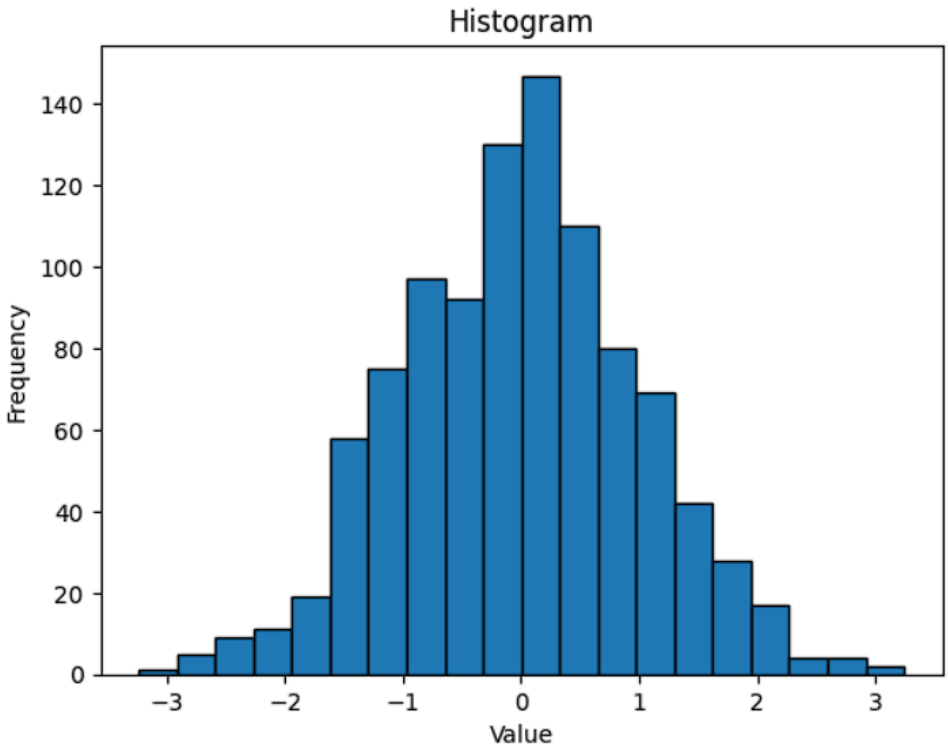
OUTPUT:

Dataset:

	Category	Values
0	A	3
1	B	7
2	C	5
3	D	4




Bar Chart
Values
Category
A
B
C
D
0
1
2
3
4
5
6
7


Histogram
Frequency
Value
0
20
40
60
80
100
120
140
−3
−2
−1
0
1
2
3